SPARQL

1) **Quelles les modalités d'APA (fréquence, intensité) pour l'activité physique renforcement musculaire ?**

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?prop ?modalite
WHERE {
 :RenforcementMusculaire rdfs:subClassOf* ?parent .

 ?parent (rdfs:subClassOf|owl:equivalentClass) ?anonClass .
 ?anonClass owl:intersectionOf ?list .
 ?list rdf:rest*/rdf:first ?element .

 ?element a owl:Restriction ;
        owl:onProperty ?prop ;
        owl:hasValue ?modalite .

 FILTER(?prop IN (:aPourFrequence, :aPourIntensite))
 }
```

SPARQL query:

```
WHERE {
 :RenforcementMusculaire rdfs:subClassOf* ?parent .

 ?parent (rdfs:subClassOf|owl:equivalentClass) ?anonClass .
 ?anonClass owl:intersectionOf ?list .
 ?list rdf:rest*/rdf:first ?element .

 ?element a owl:Restriction ;
        owl:onProperty ?prop ;
        owl:hasValue ?modalite .

 FILTER(?prop IN (:aPourFrequence, :aPourIntensite))
}
```

prop	modalite
aPourFrequence	"par semaine"
aPourIntensite	Borg modéré

2) **Quelles activités physiques sont interdites dans le cadre d'une cardiopathie inflammatoire ?**

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>
```

```
SELECT DISTINCT ?activite
WHERE {
  :CardiopathiesInflammatoires rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list ] .
  ?list rdf:rest*/rdf:first ?element .
  ?element a owl:Restriction ;
        owl:onProperty :aPourContreIndication ;
        owl:someValuesFrom ?activite .
}
```

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?activite
WHERE {
  :CardiopathiesInflammatoires rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list ] .
  ?list rdf:rest*/rdf:first ?element .
  ?element a owl:Restriction ;
        owl:onProperty :aPourContreIndication ;
        owl:someValuesFrom ?activite .
}
```

activite
Activité physique

3) Quelles activités physiques sont recommandées pour un angor stable ?

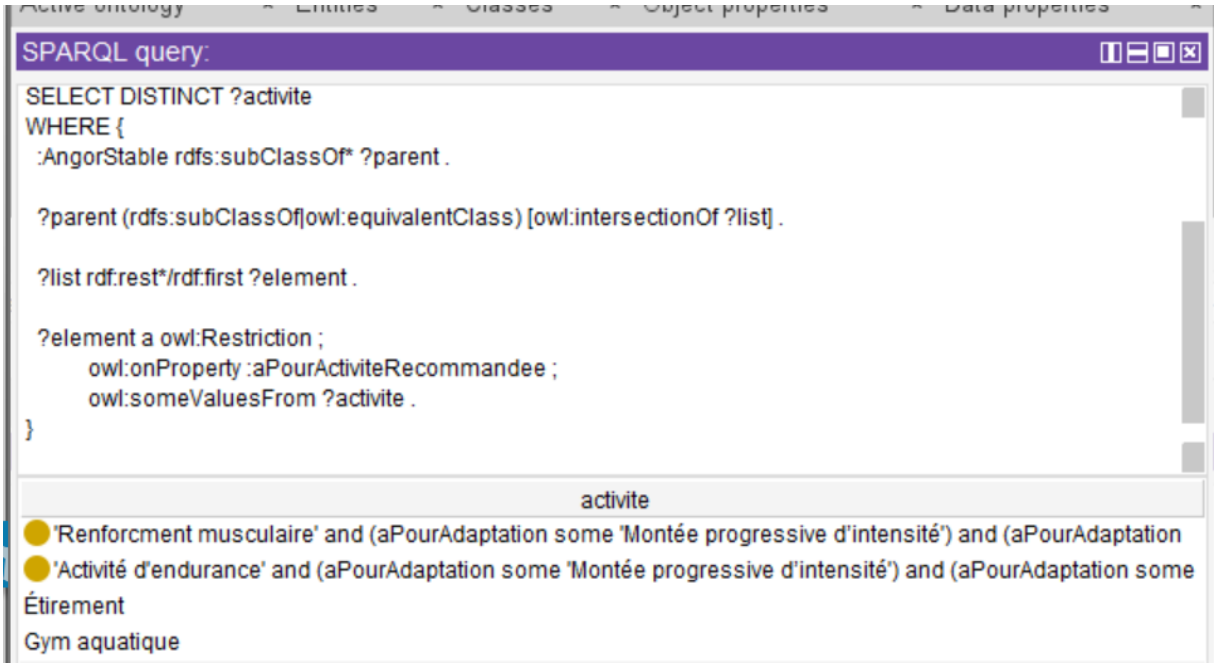
```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?activite
WHERE {
  :AngorStable rdfs:subClassOf* ?parent .

  ?parent (rdfs:subClassOf|owl:equivalentClass) [owl:intersectionOf ?list] .

  ?list rdf:rest*/rdf:first ?element .

  ?element a owl:Restriction ;
        owl:onProperty :aPourActiviteRecommandee ;
        owl:someValuesFrom ?activite .
}
```



4) Quels leviers proposer à un patient ne disposant pas d'un gymnase près de chez lui ?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?levier
WHERE {
  :AbsenceGymnase rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
  ?anon owl:intersectionOf ?list .
  ?list rdf:rest*/rdf:first ?element .
  ?element a owl:Restriction ;
        owl:onProperty :aPourLevier ;
        owl:someValuesFrom ?unionClass .
  ?unionClass owl:unionOf ?uList .
  ?uList rdf:rest*/rdf:first ?levier .
}
```

SPARQL query:

```
SELECT DISTINCT ?levier
WHERE {
 :AbsenceGymnase rdfs:subClassOf* ?parent .
 ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
 ?anon owl:intersectionOf ?list .
 ?list rdf:rest*/rdf:first ?element .
 ?element a owl:Restriction ;
        owl:onProperty :aPourLevier ;
        owl:someValuesFrom ?unionClass .
 ?unionClass owl:unionOf ?uList .
 ?uList rdf:rest*/rdf:first ?levier .
}
```

levier
Activité à domicile
Activité sans matériel
Marche autour du domicile
Micro séance 10 min

5) Quelles contre-indications d'activité physique sont associées à un syndrome coronarien aigu chez une patiente présentant également une hypertension artérielle et un stent ?

```
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl:  <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX :     <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?profil ?contreIndication
WHERE {
  VALUES ?profil {
    :SyndromeCoronarienAigu
    :Hypertension_arterielle
  }

  ?profil rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
  ?anon owl:intersectionOf ?list .
  ?list rdf:rest*/rdf:first ?restriction .

  ?restriction a owl:Restriction ;
            owl:onProperty :aPourContreIndication ;
            owl:someValuesFrom ?contreIndication .
}
```

SPARQL query:

```
  :Hypertension_arterielle
 }

 ?profil rdfs:subClassOf* ?parent .
 ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
 ?anon owl:intersectionOf ?list .
 ?list rdf:rest*/rdf:first ?restriction .

 ?restriction a owl:Restriction ;
         owl:onProperty :aPourContreIndication ;
         owl:someValuesFrom ?contreIndication .
}
```

profil	contreIndication
Syndrome coronarien aigu	Activité physique
Hypertensiopon arterielle	Travail en isométrie
Hypertensiopon arterielle	Musculation lourde

6) Quels équipements sont nécessaire pour un exercice d'abduction de la jambe ?

```
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl:  <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd:  <http://www.w3.org/2001/XMLSchema#>
PREFIX :     <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?equipement
WHERE {
  :AbductionDeJambe rdfs:subClassOf* ?parent .

  ?parent (rdfs:subClassOf|owl:equivalentClass) [
    owl:intersectionOf ?list
  ] .

  ?list rdf:rest*/rdf:first ?element .

  ?element owl:onProperty :aBesoinDe ;
          owl:someValuesFrom ?unionClasse .

  ?unionClasse owl:unionOf ?unionList .
  ?unionList rdf:rest*/rdf:first ?equipement .
}
```

SPARQL query:

```
 ?parent (rdfs:subClassOf|owl:equivalentClass) [
  owl:intersectionOf ?list
 ] .

 ?list rdf:rest*/rdf:first ?element .

 ?element owl:onProperty :aBesoinDe ;
       owl:someValuesFrom ?unionClasse .

 ?unionClasse owl:unionOf ?unionList .
 ?unionList rdf:rest*/rdf:first ?equipement .
}
```

equipement
Élastique
Machine guidée
Poulie

7) **Quelles activités physiques nécessitent une adaptation dans le cadre de la Coronaropathie fonctionnelle ?**

```
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl:  <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd:  <http://www.w3.org/2001/XMLSchema#>
PREFIX :    <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?actFiller ?adaptation
WHERE {
  :CoronaropathieFonctionnelle rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) ?ce .
  ?ce owl:intersectionOf ?list .
  ?list rdf:rest*/rdf:first ?elemAct .

  ?elemAct a owl:Restriction ;
        owl:onProperty :aPourActiviteRecommandee ;
        owl:someValuesFrom ?actFiller .

  ?actFiller owl:intersectionOf ?actList2 .
  ?actList2 rdf:rest*/rdf:first ?adRest .

  ?adRest a owl:Restriction ;
       owl:onProperty :aPourAdaptation .

  {
   ?adRest owl:someValuesFrom ?adaptation .
   FILTER NOT EXISTS { ?adaptation owl:unionOf ?u }
```

```
  }
  UNION
  {
    ?adRest owl:someValuesFrom ?uClass .
    ?uClass owl:unionOf ?u .
    ?u rdf:rest*/rdf:first ?adaptation .
  }
}
ORDER BY ?actFiller ?adaptation
```

SPARQL query:

```
  {
    ?adRest owl:someValuesFrom ?adaptation .
    FILTER NOT EXISTS { ?adaptation owl:unionOf ?u }
  }
  UNION
  {
    ?adRest owl:someValuesFrom ?uClass .
    ?uClass owl:unionOf ?u .
    ?u rdf:rest*/rdf:first ?adaptation .
  }
}
ORDER BY ?actFiller ?adaptation
```

actFiller	adaptation
'Renforcment musculaire' and (aPourAdaptation some	Montée progressive d'intensité
'Renforcment musculaire' and (aPourAdaptation some	Réduction de charge
'Activité d'endurance' and (aPourAdaptation some 'Mor	Borg faible
'Activité d'endurance' and (aPourAdaptation some 'Mor	Borg modéré
'Activité d'endurance' and (aPourAdaptation some 'Mor	Montée progressive d'intensité

8) Quels objectifs pour les étirements ?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?objectif
WHERE {
  :Etirement rdfs:subClassOf* ?parent .

  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
  ?anon owl:intersectionOf ?list .

  ?list rdf:rest*/rdf:first ?element .

  ?element a owl:Restriction ;
        owl:onProperty :aPourObjectif ;
        owl:someValuesFrom ?objectif .
```

}

SPARQL query:

```
WHERE {
  :Etirement rdfs:subClassOf* ?parent .

  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
  ?anon owl:intersectionOf ?list .

  ?list rdf:rest*/rdf:first ?element .

  ?element a owl:Restriction ;
        owl:onProperty :aPourObjectif ;
        owl:someValuesFrom ?objectif .
}
```

objectif
Amélioration de l'amplitude

9) Quels exercices physiques ont été prescrits à X ?

PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT ?prescription ?date ?activite
WHERE {
  ?prescription :concerne :Patient1 .
  ?prescription :aPourDate ?date .
  ?prescription :contient ?activite .
}

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT ?prescription ?date ?activite
WHERE {
  ?prescription :concerne :Patient1 .
  ?prescription :aPourDate ?date .
  ?prescription :contient ?activite .
}
```

prescription	date	activite
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	ActivitéEndurancePatient1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Renforcement musculaire Patient 1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Etirement patient 1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Sport_de_combat_Patient_1